Thomas Ratsakatika

Programme leader and machine learning researcher

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My work sits between research, engineering and policy. I spent ten years in the UK Government, including postings to British embassies and high commissions in Tanzania, Sierra Leone and DR Congo. In those roles, I led teams of technical specialists and programme managers, and secured £70m in new funding for infrastructure, climate and job-creation programmes. I returned to Cambridge in 2023 to specialise in machine learning and environmental science, leading research on old-growth forest detection and deploying an AI-based wildlife monitoring system, which has been in operational use since 2024.

Experience

University of Cambridge Sep 2023 – Present

PhD and MRes Researcher, UKRI Centre for Doctoral Training in AI for Environmental Risks (AI4ER)

- **Lead author of a study benchmarking geospatial foundation-model** embeddings (AlphaEarth, TESSERA) against Sentinel-1/2 features for old-growth forest detection in Romania's Făgăraș Mountains (212,000 ha). Embeddings performed best, but spatially buffered nested cross-validation narrowed their advantage. Open data and code.
- **Built and deployed a real-time wildlife alert system** with Fundația Conservation Carpathia. Photos from 4G camera traps pass through an AI pipeline that detects and classifies priority species (bears, wolves, lynx and wild boar) and alerts rapid intervention teams. Over 49,000 images processed and 4,850 alerts sent since May 2024.
- **Teaching:** supervised undergraduate Geography students in data analysis, biogeography and remote sensing; ran dissertation workshops on quantitative methods.

Biodiversity Challenge Funds Jun 2024 – Present

Expert Committee Member

- **Strategic and funding advice** for the Darwin Initiative, the UK's grant scheme for biodiversity and local communities.

Foreign, Commonwealth & Development Office (formerly DFID) Sep 2013 – Jul 2023

Head of the Private Sector Development Hub, British High Commission, Dar es Salaam, Tanzania *Aug 2020 – Jul 2023*

- **Led the UK's partnership with Tanzania on climate finance and private sector development,** managing a team of technical specialists and programme managers.
- **Led diplomatic engagement ahead of COP26,** after which Tanzania doubled its mitigation target, removed coal from its energy master plan and committed to two major forest declarations.
- **Secured ministerial approval for a £35m green growth and job-creation programme** that uses grant incentives to leverage larger investments and supports policy and regulatory reform.
- **Ran a pilot using AI on satellite imagery to detect mangrove loss** in the Rufiji Delta, with Tanzania's National Carbon Monitoring Centre and Omdena.

Deputy Team Leader, Economic Growth, British High Commission, Freetown, Sierra Leone *Jul 2017 – Aug 2020*

- **Pivoted the UK's post-Ebola economic partnership** from short-term recovery towards long-term growth and jobs.
- **Secured ministerial approval for Invest Salone,** a £30m programme supporting agribusiness, tourism and fisheries, and led its design, procurement and mobilisation.
- **Oversaw a US$50m public-private partnership to install and operate 100 solar mini-grids,** one of the largest of its kind in Africa at the time; advised on the off-grid regulatory framework.
- **Deputised for the Team Leader;** line-managed, recruited and mentored team members.

Private Sector Development Adviser, British Embassy, Kinshasa, DR Congo *Aug 2015 – Jul 2017*

- **Managed a £70m portfolio** of agriculture, renewable energy and public works programmes in a conflict setting.
- **Led a £50m market systems programme** in agriculture, river transport, renewable energy and financial services, reaching an estimated 979,000 people.
- **Led the redesign of a struggling £20m road programme** in the conflict-affected east, securing £5m in additional funding; construction provided temporary employment for 2,000 people.

Private Sector Development Adviser, Investment Climate Team, Glasgow, UK *Sep 2014 – Aug 2015*

- **Managed two research programmes,** delivered by the World Bank and an India-based NGO, building evidence for competition policy reform in Africa and Asia; represented the UK at conferences in Bangkok and Nairobi.

Economic Analyst, Chief Economist's Office, London, UK *Sep 2013 – Aug 2014*

- **Modelled global poverty trends, economic complexity and poverty-adjusted discount rates** in a team led by Professor Stefan Dercon; organised a seminar series with leading academics.

Aon Investment Consulting Sep 2010 – Aug 2012

Associate Consultant, London, UK

- **Built financial models across the major asset classes;** contributed to research on global market trends, including emerging market debt and the European debt crisis; administered a £240m derivative fund.

AgustaWestland (now Leonardo) Jul 2007 – Jun 2008

Aerospace Engineer (industrial placement), Yeovil, UK

- **Modelled a vibration reduction system in MATLAB/Simulink** and cracking in composite rotor blades; received a company award for automating reporting processes resulting in an 18-fold efficiency gain.

Education

Exp. 2027	**PhD** Machine Learning and Remote Sensing, University of Cambridge Thesis on old-growth forest detection using machine learning and remote sensing data. UKRI doctoral studentship in AI for environmental risks. Co-supervised across the Geography and Computer Science and Technology departments.	
2024	**MRes** Environmental Data Science, University of Cambridge Machine learning, statistics, environmental data analysis. Automated wildlife monitoring research.	
2021	**CFA charter** awarded, CFA Institute	
2013	**MPhil** Engineering for Sustainable Development, University of Cambridge	Distinction
2010	**MEng** Aerospace Engineering, University of Bath	First Class

Publications and talks

2026	**Ratsakatika, T.**, Zotta, M., Keshav, S. and Lines, E.R. Geospatial embeddings detect old-growth forests but buffered spatial validation narrows their advantage over Sentinel features. arXiv:2609.28194 (under review). doi.org/10.48550/arXiv.2609.28194
2026	**Ratsakatika, T.**, Zotta, M., Keshav, S. and Lines, E.R. Old-growth forest reference labels and model predictions for the Făgăraș Mountains, Romania (2020, 10 m), version 1.0.0 [dataset]. Zenodo. doi.org/10.5281/zenodo.22693148
Sep 2026	Old-growth forest detection with geospatial foundation models and spatial validation. Talk, Joint NSO–GfÖ Conference, Odense, Denmark.
Jun 2026	Detecting old-growth forests with geospatial foundation model embeddings and spatial validation. Talk, ML4EO: Machine Learning for Earth Observation, Exeter, UK.
Jul 2024	Real-time wildlife monitoring with AI and 4G camera traps. Talk, Energy and Environment Group, Department of Computer Science and Technology, Cambridge, UK.

Skills

- **Leadership and management:** accountable for portfolios of up to £70m; advising ministers and senior officials; team leadership, line management, recruitment and mentoring; contract negotiation and risk management in fragile and conflict-affected settings.
- **Policy and delivery:** diplomatic engagement and influencing, programme design and business cases, public-private partnerships.
- **Research software:** reproducible pipelines with Git, pytest and pre-commit; open code releases.
- **Machine learning:** Python, PyTorch, scikit-learn, XGBoost, CNN, ViT, Optuna; spatial cross-validation and spatial statistics; computer vision (e.g. MegaDetector, DeepFaune, SpeciesNet, DINOv3, Segment Anything).
- **Remote sensing:** Sentinel-1/2, Planet Labs, Google Earth Engine, geospatial foundation-model embeddings (AlphaEarth, TESSERA), rasterio, geopandas, xarray.